Tutorial 3

EC3301 - Martinmas Semester

Application: Linear Probability Model

This question makes use of data from the Global Entrepreneurship Monitor. The analysis uses a 10% random sample of the publicly-available 2022 GEM survey.

Consider the following **linear probability model** of enterpreneurial intent. The outcome variable `futsup` is a discrete variable: $= 1$ if answered yes to "Are you, alone or with others, expecting to start a new business, including any type of self-employment, within the next three years?"; and $= 0$ if answered no.

$$futsup_i = \alpha + \mathbf{perceptions}_i\beta + \mathbf{x}_i\gamma + u_i$$

There are four $\mathbf{perceptions}_i$ variables:[1]

- `knowent`: Do you know someone personally who started a business in the past 2 years?
- `opport`: In the next six months, will there be good opportunities for starting a business in the area where you live?
- `suskill`: Do you have the knowledge, skill and experience required to start a new business?
- `fearfail`: Would fear of failure prevent you from starting a business?

With the exception of `knowent`, all responses are provided on a agreement scale: 1 "Strongly disagree"; 2 "Somewhat disagree"; 3 "Neither agree nor disagree"; 4 "Somewhat agree"; and 5 "Strongly agree". For the pusposes of this exercise, the variables have been discretized: $= 1$ if answers 4 or 5; $= 0$ otherwise (called `AGREEopport`, `AGREEsuskill`, `AGREEfearfail`). The variable `knowent` has been discretized too: $= 1$ if knows 2 or more; $= 0$ if knows less than two people (called `YESknowent`).

[1] This grouping of variables comes from GEM (Source).

1. Under MLR 1-4, provide an interpretation of the $E[futsup_i|\mathbf{perceptions}_i, \mathbf{x}_i]$ for this linear probability model. **Hint:** take a look at Wooldridge (2025, sec. 7.5).
2. Provide an interpretation of the coefficient on `YESknownent` in the model (under MLR 1-4). **Hint:** the extra material on Interpreting Linear Models includes examples of linear probability models.

Here are the estimates of the initial model:

```
reg futsup YESknowent AGREEopport AGREEsuskill AGREEfearfailL age agesq female hhsize i.GEMEL
```

```
      Source |       SS           df       MS      Number of obs   =     7,863
-------------+----------------------------------   F(12, 7850)     =     87.51
       Model |  189.603865        12  15.8003221   Prob > F        =    0.0000
    Residual |   1417.4128     7,850  .180562139   R-squared       =    0.1180
-------------+----------------------------------   Adj R-squared   =    0.1166
       Total |  1607.01666     7,862  .204403035   Root MSE        =    .42493

------------------------------------------------------------------------------
      futsup | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  YESknowent |    .146443   .0102691    14.26   0.000     .1263129    .1665732
 AGREEopport |   .0325016   .0099776     3.26   0.001     .0129428    .0520604
AGREEsuskill |   .1719739   .0102192    16.83   0.000     .1519417    .1920062
AGREEfearf~L |  -.0382071   .0096642    -3.95   0.000    -.0571514   -.0192628
         age |  -.0002221   .0025458    -0.09   0.930    -.0052126    .0047683
       agesq |  -.0000593    .000031    -1.91   0.056      -.00012    1.40e-
06
      female |  -.0158988   .0096716    -1.64   0.100    -.0348578    .0030602
      hhsize |   .0051717   .0020908     2.47   0.013     .0010731    .0092703
             |
     GEMEDUC |
111. SOME..  |  -.0031441    .021811    -0.14   0.885    -.0458995    .0396113
1212. SEC..  |  -.0202976   .0180499    -1.12   0.261    -.0556803    .0150851
1316. POS..  |  -.0060303   .0175068    -0.34   0.731    -.0403483    .0282877
1720. GRA..  |  -.0884575   .0214728    -4.12   0.000    -.1305499   -.0463652
             |
       _cons |    .255382   .0519485     4.92   0.000     .1535491    .3572149
------------------------------------------------------------------------------
```

3. The variable `GEMEDU` has five categories: 0 “NONE”; 111 “SOME SECONDARY”; 1212 “SECONDARY DEGREE” (i.e. completed high school); 1316 “POST SECONDARY”

(i.e. diploma or bachelors); 1720 "GRAD EXP" (i.e. post-graduate qualification). The lowest education category (0 "NONE") includes individuals with only a primary-level education. According to the estimated model, what increases entrepreneurial intent: a higher level of self-perceived entrepreneurial skills or formal education?

4. There are four regions in this global survey: 1 "Middle East & Africa"; 2 "Asia"; 3 "Latin America & Caribbean"; and 4 "Europe & N America". Using the output, replicate the F-statistic shown below to test whether there are significant differences in entrepreneurial intent across regions. You must use the formula that uses SSR of the restricted and unrestricted model.

```
reg futsup YESknowent AGREEopport AGREEsuskill AGREEfearfailL age agesq female hhsize i.GEMEI

test 2.REGION 3.REGION 4.REGION
```

```
      Source |       SS           df       MS      Number of obs   =     7,863
-------------+----------------------------------   F(15, 7847)     =     87.54
       Model |  230.375574        15  15.3583716   Prob > F        =    0.0000
    Residual |  1376.64109     7,847  .175435337   R-squared       =    0.1434
-------------+----------------------------------   Adj R-squared   =    0.1417
       Total |  1607.01666     7,862  .204403035   Root MSE        =    .41885

------------------------------------------------------------------------------
      futsup | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  YESknowent |   .1268808   .0102064    12.43   0.000     .1068735    .1468881
 AGREEopport |   .0284691   .0099021     2.88   0.004     .0090584    .0478798
AGREEsuskill |   .1491597   .0102097    14.61   0.000      .129146    .1691735
AGREEfearf~L |  -.0368748   .0095332    -3.87   0.000    -.0555624   -.0181872
         age |  -.0000886   .0025099    -0.04   0.972    -.0050087    .0048315
       agesq |  -.0000499   .0000305    -1.64   0.102    -.0001098    9.91e-
06
      female |  -.0184331   .0095588    -1.93   0.054     -.037171    .0003048
      hhsize |   .0011619   .0021271     0.55   0.585    -.0030077    .0053315
             |
     GEMEDUC |
111. SOME..  |   .0690737    .022093     3.13   0.002     .0257656    .1123818
1212. SEC..  |   .0417662    .018297     2.28   0.022     .0058992    .0776332
1316. POS..  |   .0544408   .0178464     3.05   0.002     .0194571    .0894245
1720. GRA..  |   .0127636   .0223915     0.57   0.569    -.0311297    .0566569
             |
      REGION |
```

```
    2. Asia  |  -.0870125   .0161515    -5.39   0.000    -.1186737   -.0553514
3. Latin ..  |   .0503491   .0136125     3.70   0.000      .023665    .0770333
4. Europe..  |  -.1411053    .013628   -10.35   0.000    -.1678198   -.1143908
             |
       _cons |   .2550162   .0523513     4.87   0.000     .1523937    .3576387
------------------------------------------------------------------------------

 ( 1)  2.REGION = 0
 ( 2)  3.REGION = 0
 ( 3)  4.REGION = 0

       F(  3,  7847) =   77.47
            Prob > F =    0.0000
```

5. One reason that regional differences might exist is that each region has different attitudes to entrepreneurship. The GEM measures four local attitudes:

   - `equalinc`: In your country, most people would prefer that everyone had a similar standard of living
   - `nbgoodc`: In your country, most people consider starting a new business a desirable career choice
   - `nbstatus`: In your country, those successful at starting a new business have a high level of status and respect
   - `nbmedia`: In your country, you will often see stories in the public media about successful new businesses

   As above, each has been discretized to `AGREE*`. Using the output below, test whether significant regional differences remain after controlling for these attitudes using the output below.

```
est clear
qui reg futsup YESknowent AGREEopport AGREEsuskill AGREEfearfailL age agesq female hhsize i.
    est store unrestricted
qui reg futsup YESknowent AGREEopport AGREEsuskill AGREEfearfailL age agesq female hhsize i.
    est store restricted

etable, est(unrestricted restricted) novarlabel column(estimates) cstat(_r_se, nformat(%5.4f
```

```
-----------------------------------------------------
                            unrestricted restricted
```

```
-------------------------------------------------------
YESknowent                          (0.0102)   (0.0103)
AGREEopport                         (0.0101)   (0.0102)
AGREEsuskill                        (0.0103)   (0.0104)
AGREEfearfailL                      (0.0096)   (0.0097)
age                                 (0.0025)   (0.0025)
agesq                               (0.0000)   (0.0000)
female                              (0.0096)   (0.0097)
hhsize                              (0.0021)   (0.0021)
GEMEDUC
  111. SOME SECONDARY               (0.0221)   (0.0218)
  1212. SECONDARY DEGREE            (0.0183)   (0.0180)
  1316. POST SECONDARY              (0.0179)   (0.0175)
  1720. GRAD EXP                    (0.0224)   (0.0215)
REGION
  2. Asia                           (0.0162)
  3. Latin America & Caribbean      (0.0138)
  4. Europe & N America             (0.0138)
AGREEequalinc                       (0.0100)   (0.0101)
AGREEnbgoodc                        (0.0106)   (0.0107)
AGREEnbstatus                       (0.0107)   (0.0107)
AGREEnbmedia                        (0.0104)   (0.0105)
_cons                               (0.0533)   (0.0526)
Number of observations                 7863       7863
R-squared                              0.14       0.12
-------------------------------------------------------
```

6. Compute the adjusted F-statistic which has an asymptotic χ^2_q distribution under the null hypothesis.

7. What is the LM-statistic version of this test? Compute the statistic and state its asymptotic distribution. You will need the following information.

```
est clear
qui reg futsup YESknowent AGREEopport AGREEsuskill AGREEfearfailL age agesq female hhsize i.(
    predict residual, resid
reg residual YESknowent AGREEopport AGREEsuskill AGREEfearfailL age agesq female hhsize i.GE
```

```
      Source |       SS           df       MS      Number of obs   =     7,863
-------------+----------------------------------   F(19, 7843)     =     11.70
       Model |  38.9564912        19  2.05034164   Prob > F        =    0.0000
    Residual |   1374.2335     7,843  .175217838   R-squared       =    0.0276
```

```
-------------+----------------------------------   Adj R-squared   =    0.0252
       Total |  1413.18999     7,862  .179749426   Root MSE        =    .41859

------------------------------------------------------------------------------
    residual | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  YESknowent |  -.0186703   .0102209    -1.83   0.068    -.0387059    .0013653
 AGREEopport |  -.0025162   .0100591    -0.25   0.802    -.0222346    .0172023
AGREEsuskill |  -.0207685   .0103487    -2.01   0.045    -.0410547   -.0004823
AGREEfearf~L |   .0017505   .0095662     0.18   0.855    -.0170018    .0205029
         age |   .0000424   .0025097     0.02   0.987    -.0048773     .004962
       agesq |   9.99e-06   .0000305     0.33   0.743    -.0000498    .0000698
      female |  -.0025156     .00957    -0.26   0.793    -.0212754    .0162441
      hhsize |  -.0036203   .0021277    -1.70   0.089    -.0077912    .0005506
             |
     GEMEDUC |
111. SOME..  |   .0708581   .0220828     3.21   0.001       .02757    .1141463
1212. SEC..  |   .0605308   .0183007     3.31   0.001     .0246565    .0964051
1316. POS..  |    .059344   .0178554     3.32   0.001     .0243426    .0943453
1720. GRA..  |   .0987805   .0224304     4.40   0.000     .0548109    .1427501
             |
AGREEequal~c |    .008362   .0099947     0.84   0.403    -.0112302    .0279542
AGREEnbgoodc |  -.0133179   .0106483    -1.25   0.211    -.0341914    .0075556
AGREEnbsta~s |  -.0012222   .0107081    -0.11   0.909    -.0222129    .0197685
AGREEnbmedia |   -.000702   .0103859    -0.07   0.946    -.0210611    .0196571
             |
      REGION |
    2. Asia  |  -.0819987   .0162431    -5.05   0.000    -.1138396   -.0501579
3. Latin ..  |   .0553255   .0138024     4.01   0.000      .028269    .0823819
4. Europe..  |  -.1341225   .0137781    -9.73   0.000    -.1611311   -.1071138
             |
       _cons |  -.0005511   .0533441    -0.01   0.992    -.1051198    .1040176
------------------------------------------------------------------------------
```

8. Using the information below, do you reject the null under no regional differences in entrepreneurial intent using either the F-stat (Q. 6) or LM-stat (Q. 7)?

```
dis invchi2(3,0.95)
```

```
7.8147279
```

9. Using the output below, conduct a Chow test for whether the initial model differs by region. You just need to construct the F-statistic.

```
est clear
forvalues i = 1/4{
qui reg futsup YESknowent AGREEopport AGREEsuskill AGREEfearfailL age agesq female hhsize i.G
    est store region`i'
}

etable, est(region1 region2 region3 region4) column(estimates) novarlabel cstat(_r_b, nformat
```

	region1	region2	region3	region4
YESknowent	0.1649	0.1025	0.0814	0.1318
	(0.0218)	(0.0263)	(0.0208)	(0.0153)
AGREEopport	0.0318	0.1000	0.0242	0.0005
	(0.0225)	(0.0264)	(0.0207)	(0.0137)
AGREEsuskill	0.1293	0.1601	0.2018	0.0981
	(0.0226)	(0.0262)	(0.0223)	(0.0141)
AGREEfearfailL	-0.0290	-0.0674	-0.0188	-0.0498
	(0.0212)	(0.0232)	(0.0201)	(0.0135)
age	0.0066	0.0067	0.0007	-0.0077
	(0.0057)	(0.0060)	(0.0053)	(0.0037)
agesq	-0.0002	-0.0001	-0.0001	0.0000
	(0.0001)	(0.0001)	(0.0001)	(0.0000)
female	-0.0231	0.0150	-0.0465	-0.0125
	(0.0214)	(0.0232)	(0.0203)	(0.0135)
hhsize	-0.0000	0.0095	0.0037	0.0013
	(0.0029)	(0.0080)	(0.0057)	(0.0050)
GEMEDUC				
111. SOME SECONDARY	0.0003	-0.0724	0.1570	-0.0021
	(0.0495)	(0.0662)	(0.0436)	(0.0484)
1212. SECONDARY DEGREE	-0.1018	-0.0140	0.0816	0.0143
	(0.0431)	(0.0595)	(0.0288)	(0.0461)
1316. POST SECONDARY	-0.0848	-0.0397	0.1435	0.0182
	(0.0404)	(0.0591)	(0.0283)	(0.0461)
1720. GRAD EXP	-0.1057	0.1164	0.1407	-0.0374
	(0.0488)	(0.1163)	(0.0560)	(0.0471)
_cons	0.2785	0.0039	0.2124	0.3565
	(0.1141)	(0.1322)	(0.1045)	(0.0821)
Residual sum of squares	419.78	168.39	485.33	283.41

10. Is it reasonable to assume that the error term has a (conditional) standard normal distribution (MLR 6) in a linear probability model? Is this a problem for the Chow test in Q.9? And, if so, is there a solution? **Hint:** consider the material in Wooldridge (2025, secs. 7.5, 8.5).

Theory TRUE/FALSE

State whether each of the following is TRUE/FALSE and provide a reason.

1. The OLS estimator converges in distribution as follows:

$$\hat{\beta}_j \to_d N(\beta_j, Var(\hat{\beta}_j))$$

2. The finite sample variance estimator converges in probability to the asymptotic variance of the OLS estimator.

$$\frac{\sigma^2}{SSR_j} \to_p Var(\hat{\beta}_j)$$

3. t-stat $\overset{a}{\sim} N(0,1)$ under MLR 1-5, because $\hat{\sigma}^2 \to_p \sigma^2$.

Consider the linear regression model for questions 4-6

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + u_i$$

4. The test statistic for $H_0 : 2\beta_1 = 3\beta_2 - 5$ is,

$$t\text{-stat} = \frac{2\hat{\beta}_1 - 3\beta_2 - 5}{\sqrt{4Var(\hat{\beta}_1) - 9Var(\hat{\beta}_2)}}$$

5. The LM-stat for $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ is,

$$LM\text{-stat} = nR^2$$

where R^2 is the from the regression of y on x_1, x_2, x_3.

6. The restricted model for $H_0 : \beta_1 = 0.5,\ \beta_3 = 2\beta_2$ is,

$$z_i = \gamma_0 + \gamma_2 w_i + v_i$$

where $z_i = y_i - 0.5x_{i1}$ and $w_i = x_{i2} + 2x_{i3}$.

7. If the error variance is heteroskedastic, OLS is unbiased.

8. The following variance estimator has the same limit as the White formula (HC0). It is therefore heteroskedastic **consistent**.

$$\frac{(\sum_{i=1}^{n} \hat{r}_{ij}^2 \hat{u}_i^2)/(n-k-1)}{n^{-1} SSR_j^2}$$

9. The following regression can be used to test for heteroskedasticity:

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_{i1} + \delta_2 x_{i2} + \cdots + \delta_k x_{ik} + v_i$$

10. The following LM-stat statistic has a χ_k^2 distribution.

$$LM\text{-stat} = n \times \mathbf{R}_{\hat{u}^2}^2$$

where $\mathbf{R}_{\hat{u}^2}^2$ is the R-squared from the auxiliary regression in Q.9.

References

Wooldridge, Jeffrey M. 2025. "Introductory Econometrics: A Modern Approach."